APPROVED OPERATIONAL RUNBOOK: FXT-RUN-001

FIX Gateway Session Drop & Sequence Number Reset Runbook

Application: FX Trading Core (FXT-Core)	**Service:** FIX-Gateway-Svc	**Region:** GLOBAL (APAC/EMEA/AMER)
Environment: PROD / UAT	**Category:** Market Connectivity & FIX Protocols	**Version:** 3.4.1
Owner / Team: FX Electronic Trading APS Team	**Last Updated:** 2026-08-15	**Doc ID:** FXT-RUN-001

1. Operational Symptoms & Issue Summary

Issue Summary: Sudden termination or heartbeat timeout of institutional client FIX 4.2 / 4.4 trading sessions, leading to session disconnect, rejected orders, or Sequence Reset Gap Fill conflicts.

Detectable Symptoms & Alerts:

- Alert: 'FIX_SESSION_DOWN: Counterparty Goldman Sachs / Citi disconnected on port 9801'
- Log Error: 'MsgSeqNum too low, expecting 14502 but received 14200 (FIX Tag 34 mismatch)'
- Error: 'Heartbeat timeout after 30 seconds without TestRequest response (35=1)'
- Trader symptom: Order blotter shows client orders rejected with 'FIX Session Inactive'

2. Root Causes & Technical Diagnosis

- Network route flapping or WAN packet loss between client colocation and FXT gateway.
- Client restarted their FIX client engine without persisting sequence numbers, sending lower InSeqNum.
- Thread deadlock in QuickFIX/J message parser caused by memory heap spikes.

3. Mandatory Safety Directives (DO NOT DO)

CRITICAL OPERATIONAL WARNING: **DO NOT reset sequence numbers to 1 without explicit written email confirmation from the counterparty FIX trading desk.**
CRITICAL OPERATIONAL WARNING: **DO NOT terminate the primary FIX gateway process during active market trading hours (07:00-17:00 UTC) without failing over to standby port.**
CRITICAL OPERATIONAL WARNING: **NEVER manually edit session state files (.seqnums) while the QuickFIX daemon process is running; always stop the service first.**

4. Step-by-Step Resolution Procedures

Step 1: Inspect Active Session Status

Verify if the FIX session daemon is active or stuck in reconnection loop using PowerShell.

Command (PowerShell):

```
Get-Service -Name 'FXT-FixGateway' | Select-Object -Property Name, Status, StartType
```

Step 2: Check Tag 34 Sequence Mismatch in Session Logs

Search the live FIX engine log for the affected session compID to identify sequence gap.

Command (Bash/Shell):

```
grep -E '(MsgSeqNum|Tag 34|HeartBtInt)' /var/log/fxt/fix_session_GS_PROD.log | tail -n 25
```

Step 3: Perform Clean Session Stop

Gracefully disconnect the session before modifying state counters.

Command (PowerShell):

```
Stop-Service -Name 'FXT-FixGateway' -Force
```

Step 4: Apply Sequence Gap Reset

Set expected sequence numbers according to counterparty written specification.

Command (PowerShell):

```
& 'C:\FXT\FixEngine\bin\ResetSeqNum.exe' --session 'FIX.4.4:FXT->GS_NY' --inseq 14502 --outseq 18910
```

Step 5: Restart FIX Gateway Service

Restart the gateway service and watch for successful Logon (35=A) handshake.

Command (PowerShell):

```
Start-Service -Name 'FXT-FixGateway'
```

5. Post-Recovery Verification Procedures

V-1: Inspect FIX log for received 35=A (Logon) followed by sent 35=A with matching MsgSeqNum.

V-2: Verify heartbeat ping/pong messages (35=0) occur every 30 seconds continuously.

V-3: Confirm test order injection succeeds via synthetic test account: 'FXT-MON-TEST'.

6. Dependency-Aware Service Recovery Sequence

Ordered Sequence:
1. Database Connection Validator (Port 1521)
   ➔ 2. FXT Reference Data Cache (Redis)
   ➔ 3. FXT FIX Gateway Core Daemon
   ➔ 4. FIX Session Port Listener (Ports 9800-9850)

Explicit Stop/Escalate Gate: If Step 4 does not show 35=A handshake within 90 seconds, STOP recovery immediately. Do not attempt secondary reset. Escalate to On-Call FIX Engineering.

7. Edge Cases & Escalation Protocol

- **Edge Case:** Multiple rapid resets causing infinite logon loop: Trigger automatic port firewall block for 5 minutes.
- **Edge Case:** Client sending duplicate orders (PossDupFlag 43=Y): Engage Trade Surveillance desk immediately.

Primary Contact: APS FIX Lead (aps-fix-lead@fxtrading.internal | On-Call Bridge: +1-800-555-0191)	**Secondary / Bridge:** FX Electronic Trading Dev Team (fxt-eng-core@fxtrading.internal)
Severity / SLA: P1 Incident: 15-minute response SLA, 30-minute restoration target	**Escalation Condition:** If session remains disconnected after 2 clean restarts or counterparty reports order losses.